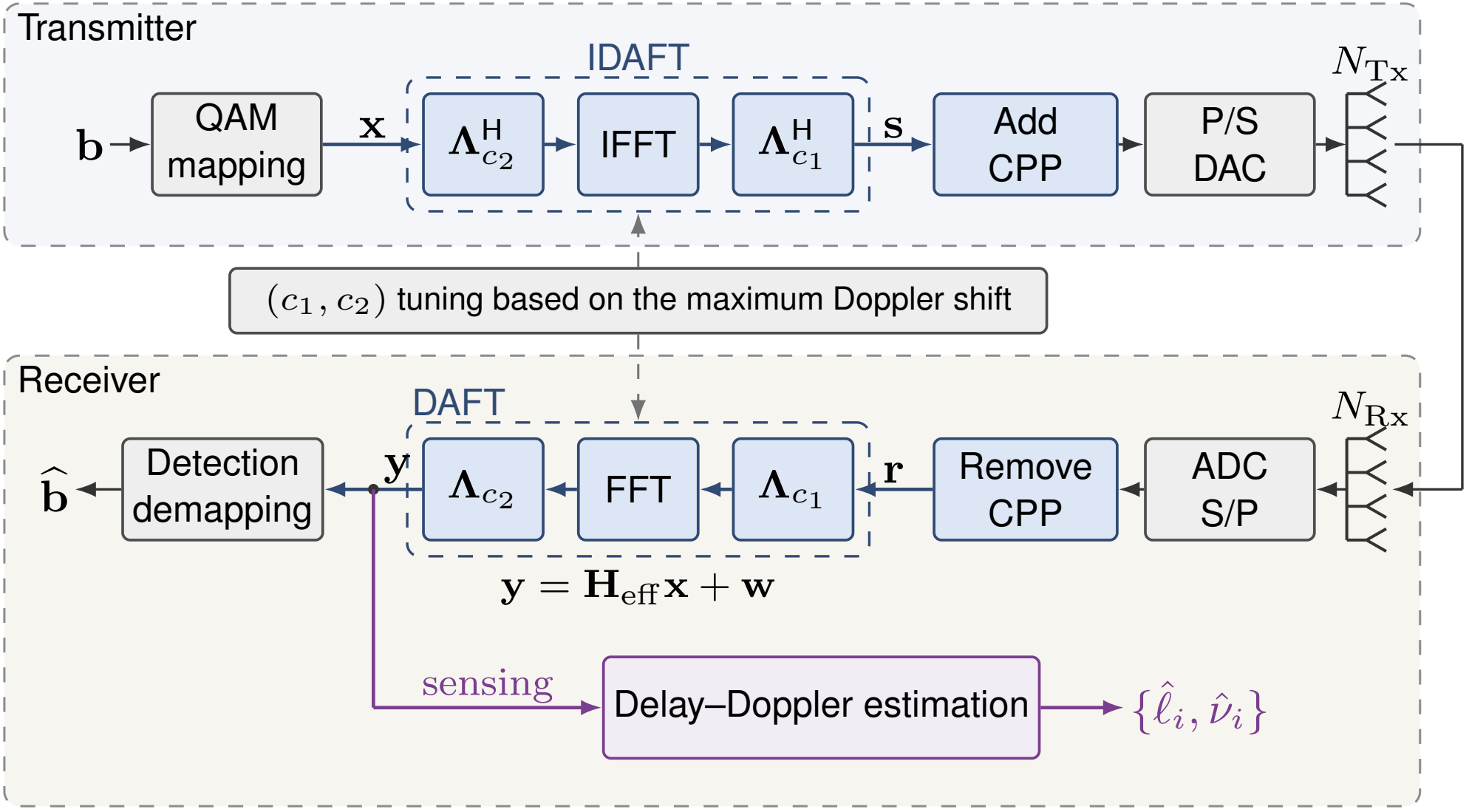

Transmitter
IDAFT
N_{Tx}
$\mathbf{b}$
QAM mapping
$\mathbf{x}$
$\mathbf{\Lambda}_{c_2}^{\mathrm{H}}$
IFFT
$\mathbf{\Lambda}_{c_1}^{\mathrm{H}}$
$\mathbf{s}$
Add CPP
P/S DAC
(c_1, c_2) tuning based on the maximum Doppler shift
Receiver
DAFT
N_{Rx}
$\widehat{\mathbf{b}}$
Detection demapping
$\mathbf{y}$
$\mathbf{\Lambda}_{c_2}$
FFT
$\mathbf{\Lambda}_{c_1}$
$\mathbf{r}$
Remove CPP
ADC S/P
$\mathbf{y} = \mathbf{H}_{\mathrm{eff}}\mathbf{x} + \mathbf{w}$
sensing
Delay–Doppler estimation
$\{\hat{\ell}_i, \hat{\nu}_i\}$